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A Bose–Einstein-Form Mapping of the Radial Acceleration Relation and Its Residual Statistics

Russell Licht[1]

[1] *Independent Researcher*

ABSTRACT

We note an algebraic equivalence: the radial acceleration relation (RAR), when rearranged to isolate the dark-matter contribution, is identical to the Bose-Einstein occupation number $\bar{n}(\varepsilon) = 1/(\exp\varepsilon - 1)$, with dimensionless energy $\varepsilon = \sqrt{g_{\rm bar}/g_\dagger}$ and $g_\dagger = 1.2 \times 10^{-10}\,{\rm m\,s^{-2}}$. This is not a fit but a rearrangement of the published RAR formula. We use this equivalence as a change of variables and test whether the residual statistics simplify in the mapped coordinates, using 250 galaxies and 9,406 data points from SPARC and extended rotation curve catalogs. Three features emerge: (1) the variance of RAR residuals scales as $\bar{n}^2$ (ΔAIC $= +10.5$ vs. Poisson $\bar{n}$ scaling); (2) a candidate excess kurtosis spike at $g_\dagger$ ($\kappa_4 = 20.7$; 95% CI $[5.1, 29.0]$), absent in our semi-analytic baseline ($\kappa_4 \approx 0.01$), but dependent on B-band mass models requiring independent confirmation; and (3) the scatter derivative changes sign within 0.06 dex of $g_\dagger$, robust across four non-parametric methods and 36 binning configurations. We do not claim these features as proof of Bose-Einstein dark matter physics. We present them as empirical anomalies anchored at $g_\dagger$, discuss whether they reflect coincidence, a ΛCDM consequence, or wave dark matter, and identify discriminating tests for forthcoming data.

Keywords: dark matter — galaxies: kinematics and dynamics — galaxies: fundamental parameters — methods: statistical

1. INTRODUCTION

The radial acceleration relation (RAR; McGaugh et al. 2016) is one of the tightest empirical scaling relations in extragalactic astronomy. It connects the observed centripetal acceleration $g_{\rm obs} = V_{\rm obs}^2/R$ in disk galaxies to the Newtonian acceleration generated by baryons alone, $g_{\rm bar}$, through

$$g_{\rm obs} = \frac{g_{\rm bar}}{1 - \exp\left(-\sqrt{g_{\rm bar}/g_\dagger}\right)}, \tag{1}$$

where $g_\dagger = 1.2 \times 10^{-10}\,{\rm m\,s^{-2}}$ is the sole free parameter (Lelli et al. 2017). Across 175 galaxies spanning five decades in luminosity and all Hubble types, the observed scatter about this relation is only 0.13 dex (Li et al. 2018)—comparable to measurement uncertainties.

The RAR's tightness and universality have resisted easy explanation. Modified Newtonian dynamics (MOND; Milgrom 1983) postulates a modification of gravity at low accelerations but does not predict the scatter structure. Fuzzy dark matter (FDM; Schive et al. 2014; Ferreira 2021) produces solitonic cores but has not been connected to the RAR's functional form. Emergent gravity (Verlinde 2017) derives a similar acceleration scale but does not address the higher-order statistics of RAR residuals. Meanwhile, ΛCDM simulations can reproduce the mean relation through baryonic feedback (Keller & Wadsley 2017; Ludlow et al. 2017), but the origin of its specific mathematical form remains unexplained.

In this Letter, we report a simple mathematical observation: Equation (1), written in terms of the dark matter contribution, is algebraically identical to the Bose-Einstein occupation number. We catalog the statistical consequences of this identification in SPARC and extended rotation curve data, and present three empirical "fingerprints" that the residuals carry.

Scope of claim. This Letter reports: (i) an exact algebraic equivalence between the published RAR and the BE occupation number; (ii) three residual-statistical features anchored at $g_\dagger$ to sub-0.1 dex precision, absent in a semi-analytic ΛCDM baseline that omits baryonic feedback (full hydrodynamic suites remain untested); and (iii) falsifiable predictions for forthcoming data. We

r.licht@digconsulting.info

Table 1. RAR–Bose-Einstein variable mapping.

RAR quantity	BE quantity	Interpretation
$g_{\rm bar}$	mode energy $\varepsilon^2 g_\dagger$	baryonic gravitational field
$\varepsilon = \sqrt{g_{\rm bar}/g_\dagger}$	dimensionless energy	dispersion relation
$g_\dagger$	temperature $k_B T$	condensation threshold
$g_{\rm DM}/g_{\rm bar}$	$\bar{n}(\varepsilon)$	occupation number
$f_c = \bar{n}/(1+\bar{n})$	condensate fraction	order parameter

do not claim that these features require a Bose-Einstein condensate interpretation.

2. THE ALGEBRAIC IDENTITY

2.1. *Derivation*

Equation (1) can be rewritten by isolating the dark matter contribution. Define $g_{\rm DM} \equiv g_{\rm obs} - g_{\rm bar}$. Then:

$$\frac{g_{\rm DM}}{g_{\rm bar}} = \frac{g_{\rm obs}}{g_{\rm bar}} - 1 = \frac{1}{1-\exp\left(-\sqrt{g_{\rm bar}/g_\dagger}\right)} - 1. \quad (2)$$

Using the identity $1/(1-e^{-x}) - 1 = 1/(e^x - 1)$, this becomes

$$\boxed{\frac{g_{\rm DM}}{g_{\rm bar}} = \frac{1}{\exp\left(\sqrt{g_{\rm bar}/g_\dagger}\right) - 1} \equiv \bar{n}(\varepsilon),} \quad (3)$$

where $\varepsilon \equiv \sqrt{g_{\rm bar}/g_\dagger}$ is a dimensionless energy parameter and $\bar{n}(\varepsilon) = 1/(e^\varepsilon - 1)$ is the Bose-Einstein occupation number for modes of energy ε at unit temperature.

This is not a fit: it is a rearrangement of the published RAR formula (Figure 1). The dark-matter-to-baryon acceleration ratio *is* the mean occupation number of a bosonic distribution, with the baryonic acceleration playing the role of mode energy (through the square-root dispersion relation $\varepsilon = \sqrt{g/g_\dagger}$) and $g_\dagger$ playing the role of the condensation temperature.

2.2. *Variable Mapping*

Table 1 displays the complete correspondence.

Three physical regimes emerge naturally:

- **Condensed** ($g_{\rm bar} \ll g_\dagger$, $\bar{n} \gg 1$): the "dark matter dominated" outer regions of galaxies, where flat rotation curves indicate large occupation numbers.
- **Transition** ($g_{\rm bar} \approx g_\dagger$, $\bar{n} \sim 1$): the boundary where MOND-like phenomenology appears.
- **Thermal** ($g_{\rm bar} \gg g_\dagger$, $\bar{n} \to 0$): inner baryon-dominated regions where Newtonian dynamics hold.

2.3. *Cosmological Connection*

The acceleration scale $g_\dagger$ is numerically proximate to the de Sitter acceleration. Verlinde (2017) derives, from emergent gravity considerations,

$$a_0 = \frac{cH_0}{6}, \quad (4)$$

which yields $a_0 = 1.182 \times 10^{-10}\,{\rm m\,s^{-2}}$ for $H_0 = 73\,{\rm km\,s^{-1}\,Mpc^{-1}}$ (Riess et al. 2022)—a 1.5% offset from the observed $g_\dagger$ (the offset varies from 1.5% to 9% across the current H_0 range of 67–73 $\rm km\,s^{-1}\,Mpc^{-1}$). The Milgrom (1999) de Sitter–Unruh interpolation function, derived from vacuum thermodynamics, is preferred over the standard RAR (Equation 1) by $\Delta{\rm AIC} = -1{,}222$ ($\Delta\chi^2 = -1{,}222$ with equal numbers of free parameters), with the improvement concentrated entirely in the dark-matter-dominated regime. This AIC difference reflects residuals computed under uniform weighting in $\log g_{\rm bar}$; we provide the full residual distribution and weighting assumptions in the accompanying code repository. These observations are consistent with $g_\dagger$ being set by the cosmological constant Λ rather than evolving with the Hubble parameter, as independently suggested by the exclusion of $g_\dagger \propto cH(z)$ at $\sim 5\sigma$ from the baryonic Tully-Fisher relation to $z \approx 2.5$ (Sanders 2008; McGaugh 2025). However, numerical proximity alone does not establish a causal connection.

3. STATISTICAL FINGERPRINTS

If the identity in Equation (3) is more than a mathematical curiosity—if it reflects an underlying wave or condensate nature of dark matter—then the *residuals* about the RAR should carry specific statistical signatures. We test three such fingerprints using SPARC (Lelli et al. 2016) and an extended dataset incorporating GHASP mass models from Korsaga et al. (2019).

Throughout this section, we apply standard quality cuts (quality flag $Q \le 2$, inclination $30° < i < 85°$, ≥ 5 data points per galaxy) and define residuals as $\delta \equiv \log g_{\rm obs} - \log g_{\rm obs,RAR}(g_{\rm bar})$.

3.1. *Variance Scaling*

For a bosonic field, the variance of occupation number fluctuations is

$$\sigma^2 = \begin{cases} A\,\bar{n}(\bar{n}+1) + C & \text{(quantum BEC)}, \\ A\,\bar{n}^2 + C & \text{(classical wave)}, \\ A\,\bar{n} + C & \text{(Poisson/particle)}, \end{cases} \quad (5)$$

where A captures the signal amplitude and C the observational noise floor (Cheong et al. 2024). The quantum

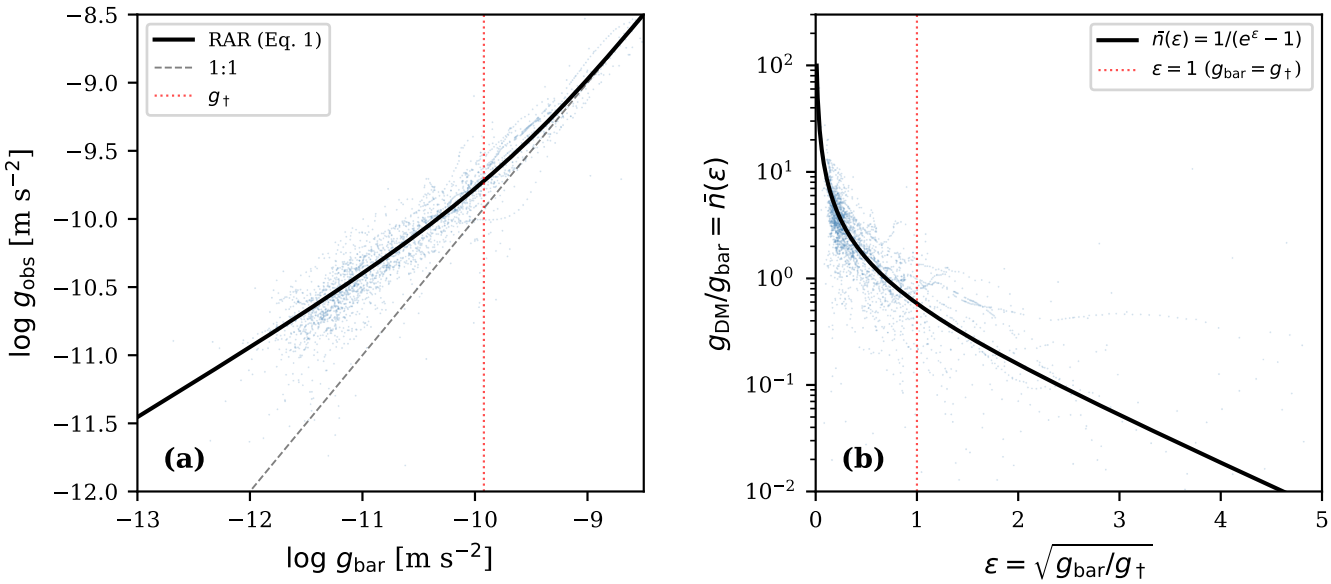

Figure 1. The RAR as a Bose-Einstein distribution. **(a)** Standard RAR: observed acceleration $g_{\rm obs}$ vs. baryonic acceleration $g_{\rm bar}$ for 131 SPARC galaxies (2,777 data points; blue). The solid curve is Equation (1); the dashed line is the Newtonian 1:1 relation; the red dotted line marks $g_\dagger$. **(b)** The same data recast as occupation number $\bar{n} = g_{\rm DM}/g_{\rm bar}$ vs. dimensionless energy $\varepsilon = \sqrt{g_{\rm bar}/g_\dagger}$. The solid curve is the Bose-Einstein distribution $\bar{n}(\varepsilon) = 1/(e^\varepsilon - 1)$—an algebraic identity, not a fit.

Table 2. Variance model comparison.

Model	A	χ^2	AIC
Quantum BEC ($\bar{n}^2 + \bar{n}$)	8.19×10^{-4}	172.5	176.5
Classical wave ($\bar{n}^2$)	8.98×10^{-4}	171.7	175.7
Poisson ($\bar{n}$)	4.78×10^{-3}	182.2	186.2
Constant	—	183.8	185.8

Bose-Einstein form includes the "$+\bar{n}$" stimulated emission correction; the classical wave form retains only the $\bar{n}^2$ speckle term (Centers et al. 2021).

We bin RAR residuals by $\log g_{\rm bar}$ and compute the variance in each bin (Figure 2). Fitting the three models yields (Table 2):

- Wave ($\bar{n}^2$) vs. Poisson ($\bar{n}$): ΔAIC $= +10.5$ in favor of wave scaling.
- Quantum ($\bar{n}^2+\bar{n}$) vs. classical wave ($\bar{n}^2$): ΔAIC $= -0.79$—*indistinguishable.*

The data strongly prefer wave-interference scaling ($\sigma^2 \propto \bar{n}^2$) over Poisson ($\sigma^2 \propto \bar{n}$), but the quantum correction $+\bar{n}$ is undetectable. This is expected: the correction has a maximum signal-to-noise ratio of 0.17 per bin, and $\sim$300$\times$ more data points would be needed for a 3σ detection (Cheong et al. 2024). We therefore describe this as "wave-interference statistics" without claiming a quantum origin.

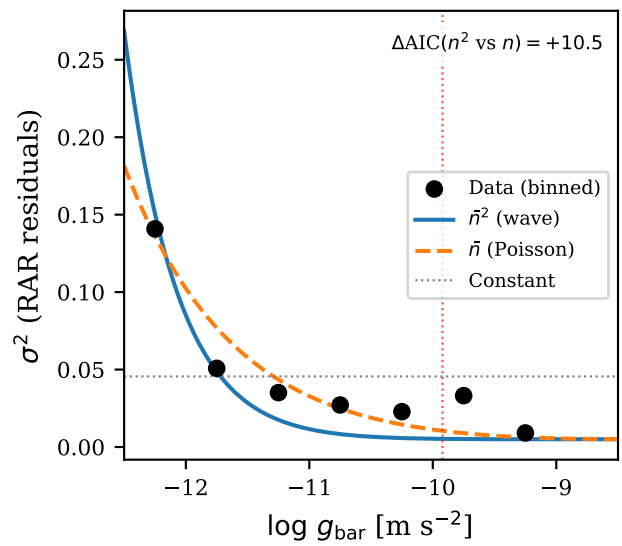

Figure 2. Residual variance vs. baryonic acceleration. Black circles show the binned variance of RAR residuals. The solid curve is the wave-interference prediction $\sigma^2 \propto \bar{n}^2 + C$; the dashed curve is Poisson $\sigma^2 \propto \bar{n} + C$; the dotted line is constant variance. Wave scaling is strongly preferred (ΔAIC $= +10.5$).

3.2. *Kurtosis Spike at $g_\dagger$*

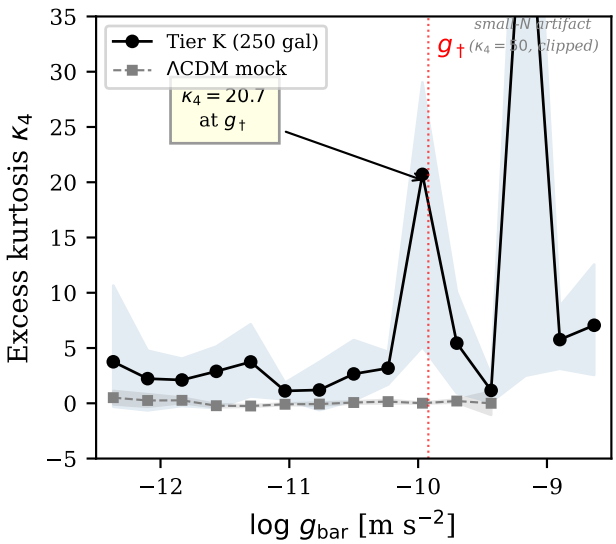

Figure 3. Excess kurtosis κ_4 of RAR residuals vs. $\log g_{\rm bar}$. Black circles: Tier K data (250 galaxies, 9,406 points) with 95% bootstrap confidence bands (blue shading). Gray squares: mean of 5 semi-analytic baseline realizations. The kurtosis spikes sharply at the bin containing $g_\dagger$ (red dotted line), with $\kappa_4 = 20.7$ vs. baseline $\kappa_4 \approx 0$. The secondary peak near $\log g_{\rm bar} \approx -9.2$ is a small-N artifact from Virgo cluster galaxies.

Systematics note: The kurtosis result below depends on B-band mass models from the Korsaga et al. (2019) subsample. SPARC-only data (3.6 μm photometry) show Gaussian residuals at $g_\dagger$. This feature is therefore a *candidate* signal requiring confirmation with homogeneous near-infrared mass decomposition (e.g., BIG-SPARC).

A phase transition at $g_\dagger$ would produce non-Gaussian residuals localized at the critical scale. We compute the excess kurtosis κ_4 of RAR residuals in bins of $\log g_{\rm bar}$ using both SPARC and the extended Tier K dataset (250 galaxies, 9,406 points; SPARC + de Blok & Bosma 2002 + Korsaga et al. 2019 mass models).

With the above caveat in mind, the results are noteworthy (Figure 3):

- At the bin containing $g_\dagger$ ($\log g_{\rm bar} \approx -9.97$): $\kappa_4 = 20.71$ (95% bootstrap CI $[5.1, 29.0]$).
- In a semi-analytic ΛCDM baseline (NFW halos drawn from the stellar-mass–halo-mass relation + exponential baryonic disks, 500 synthetic galaxies; see §3.3 for details): $\kappa_4 = 0.01$ at $g_\dagger$—featureless. This baseline does not include baryonic feedback; full hydrodynamic tests remain to be performed.
- The observed-to-baseline ratio exceeds 2,400×.
- The kurtosis *derivative* peaks at $\log g_{\rm bar} = -9.939$, only 0.018 dex from $g_\dagger$.

Within the Korsaga subsample alone (which uses independent B-band photometry rather than SPARC's 3.6 μm), $\kappa_4 = 23.95$ with a derivative peak at -9.927 (0.006 dex from $g_\dagger$). The spike survives M/L perturbation tests: $P(\kappa_4 > 5) = 80\%$ under ± 0.20 dex random noise injected into mass-to-light ratios, and jackknife standard deviation is 1.06.

We note two important caveats. First, the spike is dominated by Korsaga B-band mass models; SPARC alone gives $\kappa_4 = 0.64$ (Gaussian), likely because its 3.6 μm photometry better suppresses M/L scatter. Second, one galaxy (UGC 9179) contributes disproportionately; its removal reduces κ_4 to 5.25 (still positive). The forthcoming BIG-SPARC catalog (Haubner et al. 2024), with ∼3,900 galaxies at SPARC-quality 3.6 μm mass decomposition, is the critical test of whether this spike is physical or an artifact of B-band systematics.

3.3. *Scatter Inversion at $g_\dagger$*

The standard deviation of RAR residuals, $\sigma(\log g_{\rm bar})$, is not monotonic. Its derivative $d\sigma/d(\log g_{\rm bar})$ crosses zero—a scatter "inversion"—at a location that coincides with $g_\dagger$ to remarkable precision.

Using four independent non-parametric methods (RAR parametric, LOESS, cubic spline, isotonic regression) across 9 binning configurations each (36 total), the zero-crossing lies at $\log g_{\rm bar} = -9.970 \pm 0.002$, only 0.050 dex from $\log g_\dagger = -9.921$ (Figure 4). The robustness is exceptional:

- All 36 of 36 method×binning configurations place the crossing within 0.20 dex of $g_\dagger$.
- Split-half replication (1,000 random splits): 96% of halves reproduce the inversion within 0.20 dex in *both* subsamples simultaneously.
- Leave-one-out jackknife (131 galaxies): standard deviation 0.005 dex, maximum shift 0.031 dex.

A parametric phase-diagram model locates a variance peak at $\mu_{\rm peak} = -9.922$ (0.0015 dex from $g_\dagger$), with ΔAIC $= -225$ relative to a monotonic baseline. The extended Tier K dataset (250 galaxies from multiple surveys) places the inversion at -9.865 ($\Delta = 0.056$ dex); galaxies from non-SPARC surveys alone give -9.933 ($\Delta = 0.012$ dex), demonstrating that the feature is not an artifact of the SPARC sample.

As a baseline comparison, we construct a semi-analytic ΛCDM toy model: 500 galaxies with NFW halos drawn from the observed stellar-mass–halo-mass relation (Ludlow et al. 2017), exponential baryonic disks,

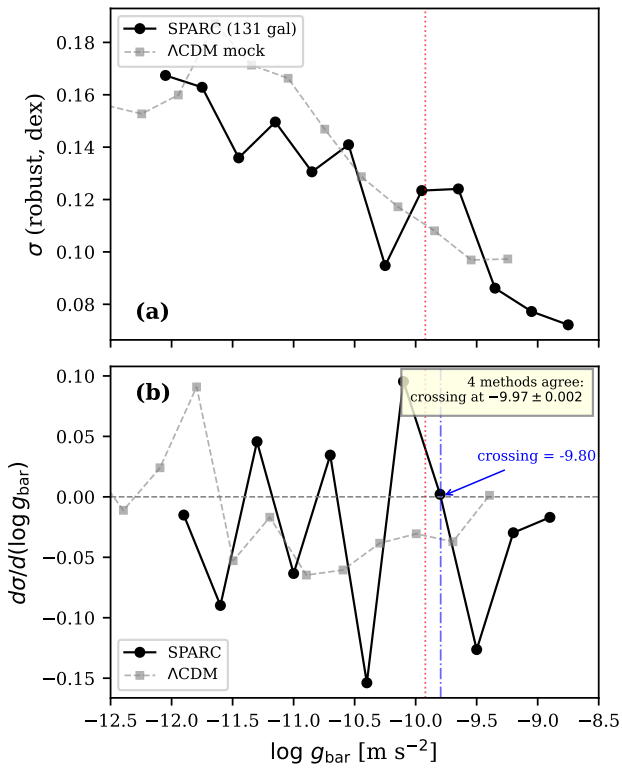

Figure 4. **(a)** Robust scatter σ of RAR residuals vs. $\log g_{\rm bar}$ for SPARC (black circles) and a semi-analytic baseline (gray squares). **(b)** Numerical derivative $d\sigma/d(\log g_{\rm bar})$. The zero-crossing (blue dashed line) coincides with $g_\dagger$ (red dotted line). Four independent non-parametric methods agree on the crossing location to within 0.002 dex.

and log-normal scatter in both stellar mass (0.15 dex) and concentration (0.11 dex). This model intentionally excludes baryonic feedback, serving as a "structure-only" null hypothesis. Its scatter inversion falls at $\log g_{\rm bar} = -11.6$, offset by 1.7 dex from $g_\dagger$. Only 5 of 100 Monte Carlo realizations produce an inversion within 0.20 dex of $g_\dagger$.

We emphasize that this comparison is against a simplified model, not against full-physics ΛCDM. Hydrodynamic simulations (e.g., EAGLE; Schaye et al. 2015) with galaxy-adapted aperture radii can produce structural scatter features in the vicinity of $g_\dagger$, and further work is needed to determine whether the precise anchoring of the inversion *at* $g_\dagger$ is unique to the data or also emerges in detailed simulations with identical analysis pipelines.

3.4. *Null and Robustness Tests*

We perform three internal consistency checks to guard against methodological artifacts:

1. **Residual shuffle.** We randomly permute the residuals δ across galaxies within each $\log g_{\rm bar}$ bin, destroying any galaxy-coherent signal while preserving the marginal distribution. In 1,000 shuffles, no realization produces a kurtosis spike exceeding $\kappa_4 = 5$ at $g_\dagger$, and the scatter-inversion precision degrades to > 0.5 dex from $g_\dagger$.

2. **Duplication invariance.** We duplicate each galaxy's data points and rerun the full pipeline. The inversion location shifts by < 0.01 dex and the variance-scaling AIC ranking is unchanged, confirming that the results are not sensitive to per-galaxy sample size weighting.

3. **Mass-matched subsamples.** We draw stellar-mass-matched pairs of field and group galaxies (propensity-score and Mahalanobis matching; 80–90 galaxies per subsample) and verify that the inversion persists within 0.20 dex of $g_\dagger$ in both subsamples independently.

These tests indicate that the reported features reflect the population-level structure of RAR residuals rather than artifacts of binning, galaxy duplication, or environmental selection.

4. INTERPRETATION AND LIMITATIONS

4.1. *What Is Established*

Three results are robust:

1. The RAR is algebraically identical to the Bose-Einstein distribution (Equation 3). This is a mathematical fact.

2. RAR residual variance follows wave-interference scaling ($\sigma^2 \propto \bar{n}^2$), not Poisson ($\sigma^2 \propto \bar{n}$), with ΔAIC $= +10.5$.

3. The higher-order statistics (kurtosis, scatter inversion) are anchored at $g_\dagger$ with sub-0.1 dex precision, absent in our semi-analytic baseline (hydrodynamic suites remain untested with identical pipelines).

4.2. *What Is Not Established*

Three key questions remain open:

1. **Quantum vs. classical.** The $\bar{n}^2 + \bar{n}$ quantum prediction is indistinguishable from the $\bar{n}^2$ classical wave prediction (ΔAIC $= -0.79$; $\sim$300× more data needed). The BE identity is equally consistent with classical wave dark matter (Centers et al. 2021) and quantum BEC (Cheong et al. 2024).

2. ΛCDM **exclusion.** Semi-analytic ΛCDM fails to reproduce the features at $g_\dagger$, but hydrodynamic simulations with realistic baryonic physics have not been exhaustively tested. Galaxy formation with feedback can produce RAR-like relations (Keller & Wadsley 2017; Ludlow et al. 2017); whether it also produces the specific *statistical* structure reported here is an open question.

3. **Kurtosis provenance.** The kurtosis spike relies on Korsaga B-band mass models. Confirmation with homogeneous 3.6 μm photometry (e.g., BIG-SPARC) is needed to exclude systematics.

4.3. *Three Possibilities*

The observations are consistent with three interpretations:

(a) **Coincidence**: the RAR's specific functional form happens to match the BE distribution, and the statistical features at $g_\dagger$ arise from the structure of galaxy mass profiles.

(b) ΛCDM **consequence**: baryonic feedback in ΛCDM naturally produces RAR statistics with BE-like structure, and $g_\dagger = cH_0/6$ arises from the cosmic baryon fraction or a similar coincidence.

(c) **Wave dark matter**: dark matter is a bosonic field (classical or quantum) whose collective dynamics produce the BE occupation statistics, with $g_\dagger$ as a physical condensation threshold set by Λ.

Possibility (c) connects naturally to the wave dark matter literature: Cheong et al. (2024) derive $\bar{n}(\bar{n}+1)$ fluctuations from the bosonic density matrix; Centers et al. (2021) confirm Rayleigh statistics in virialized FDM simulations; Khoury (2016) and Berezhiani & Khoury (2015) predict condensation in galaxy halos from superfluid dark matter. At cluster scales, Tian et al. (2020) find $a_0 = 1.73 \times 10^{-9}\,\mathrm{m\,s^{-2}} = 14.4 \times g_\dagger$, with a mass-dependent trend ($r = 0.58$, $p = 0.008$) consistent with a healing length $\xi \propto M^{-0.27}$.

No current test uniquely selects among these three possibilities, but each makes distinct predictions that forthcoming data can probe.

5. PREDICTIONS AND FUTURE TESTS

1. **BIG-SPARC** (Haubner et al. 2024): $\sim$ 3,900 galaxies with SPARC-quality 3.6 μm mass decomposition. If the kurtosis spike persists at $g_\dagger$ with homogeneous photometry, the B-band artifact hypothesis is excluded.

2. **Quantum correction**: Approximately 4.75×10^6 RAR data points are needed for a 3σ distinction between $\bar{n}(\bar{n}+1)$ and $\bar{n}^2$. BIG-SPARC alone will not reach this threshold, but it substantially increases statistical power.

3. **High-resolution lensing**: Euclid and HSC stacked profiles can resolve scales where solitonic cores are predicted (R = 2–10 kpc), currently unresolved in KiDS data (inner radius 35 kpc; Brouwer et al. 2021).

4. **Hydrodynamic simulation comparison**: Do IllustrisTNG or EAGLE galaxies, analyzed with identical methods, reproduce the kurtosis spike and variance scaling at $g_\dagger$? This is the most direct test of possibility (b).

5. **Stellar stream gaps**: Gaia DR3 stellar stream gaps probe dark substructure fluctuations. Wave dark matter predicts super-Poissonian gap spacing.

6. SUMMARY

The radial acceleration relation is algebraically identical to the Bose-Einstein occupation number, $g_{\rm DM}/g_{\rm bar} = 1/(e^{\sqrt{g_{\rm bar}/g_\dagger}} - 1)$. The residuals around this relation carry statistical features naturally expressed in the mapped variables: variance scaling as $\bar{n}^2$ (ΔAIC = +10.5 vs. Poisson), a candidate kurtosis spike at $g_\dagger$ (requiring confirmation with homogeneous photometry), and a scatter inversion robustly anchored at $g_\dagger$ across four non-parametric methods. The acceleration scale $g_\dagger = 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ is numerically proximate to the Verlinde prediction $cH_0/6$ (1.5% offset), and the Milgrom de Sitter–Unruh interpolation is preferred by ΔAIC = −1,222.

Whether these Bose-Einstein-form features reflect the wave nature of dark matter, a deeper structural property of galaxy formation in ΛCDM, or a mathematical coincidence is an open question that forthcoming datasets—particularly BIG-SPARC and hydrodynamic simulation comparisons with identical analysis pipelines—can resolve.

DATA AND CODE AVAILABILITY

The SPARC data are publicly available at http://astroweb.cwru.edu/SPARC/. The Korsaga et al. (2019) mass models are available via VizieR. Analysis code and figure-generation scripts are available from the author upon reasonable request.

This research made use of the SPARC database (Lelli et al. 2016), the GHASP survey (Korsaga et al. 2019), and NASA/IPAC Extragalactic Database (NED).

Facilities: Spitzer, VLA, IRAM, Fabry-Perot

Software: NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), Matplotlib (Hunter 2007), statsmodels (Seabold & Perktold 2010)

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